
KALIBRATE: AN AGENTIC FORECASTING SYSTEM FOR POP-CULTURE PREDICTION MARKETS

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ABSTRACT

We present KALIBRATE, an agentic forecasting system for producing calibrated probability estimates on pop-culture prediction markets. The system combines a structured market-state constructor, budgeted web-search routing, evidence extraction, schema-constrained probability forecasting, and a risk controller that can abstain under high uncertainty. We evaluate KALIBRATE on POPCAST, a 10-task benchmark of historical pop-culture forecasting scenarios spanning film, television, music, awards, celebrity activity, streaming, and viral media. Across 130 total evaluation runs, the full KALIBRATE pipeline achieves a mean Brier score of 0.173 and task-success rate of 0.800, beating a random 0.5 prior but underperforming all non-random baselines: a single-prompt frontier LLM (0.101), a tool-enabled frontier LLM (0.112), and especially our own market-only ablation (0.076). This counter-intuitive result is the central finding of the paper. Market priors and task context already contain strong signal, while retrieval can actively hurt in noisy entertainment domains by amplifying rumors, hype, and temporal leakage risk. We analyze this negative result through difficulty, domain, and per-task breakdowns, identifying celebrity and music tasks as the primary failure modes. Our conclusion is not that agentic forecasting is unpromising, but that pop-culture forecasting requires auditable evidence, finer-grained source credibility, calibrated abstention, and better decisions about when not to retrieve.

1 INTRODUCTION

Prediction markets ask participants to translate uncertain information into prices. On platforms such as Kalshi, a contract may ask whether a film will cross a box-office threshold, whether an artist will receive an award nomination, or whether a public figure will announce a major life event before a deadline. The price of a YES contract is an implied probability, and good forecasters must decide whether the market is underpricing or overpricing the event.

Pop-culture markets are a useful stress test for language agents because the relevant evidence is both public and unreliable. Forecasts depend on box-office tracking, award-season history, chart momentum, social media virality, leaks, entertainment reporting, and rumor cycles. An agent must decide not only what evidence to retrieve, but whether that evidence deserves to move the prior at all. This is a different task from static question answering: the goal is not to recover a known answer, but to produce a calibrated probability under a historical information cutoff.

We built KALIBRATE to study this problem. KALIBRATE is a modular forecasting agent that receives a structured market state, decides whether to retrieve external evidence, converts retrieved documents into typed evidence features, forecasts a YES probability, and optionally abstains by returning a neutral probability of 0.5. The agent is evaluated using proper scoring rules rather than only directional accuracy.

Our original hypothesis was that this structured agentic workflow would outperform both one-shot prompting and raw tool use, especially on hard tasks where external evidence should matter. The final results refute that hypothesis. The full agent beats a random prior but loses to all non-random baselines, including a market-only ablation of itself. Rather than treating this as a failed experiment,

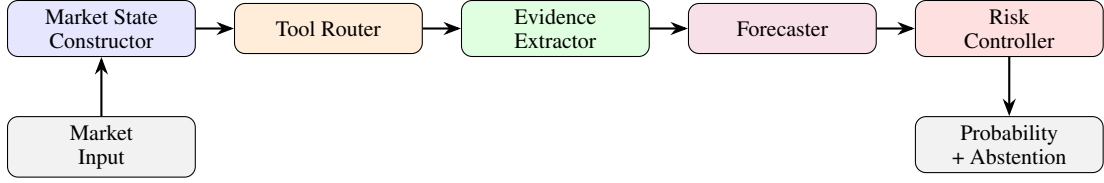


Figure 1: KALIBRATE architecture. The agent converts a market into structured state, optionally retrieves evidence under a budget, extracts typed signals, forecasts a probability, and may abstain under high uncertainty.

we treat it as the main scientific result: in our benchmark, market prices and curated task context were often safer than live retrieval, and retrieval failures were concentrated in domains where the web over-produces plausible but low-reliability signals.

Our contributions are:

1. A complete agentic forecasting pipeline for pop-culture prediction markets, including market-state construction, budgeted tool routing, structured evidence extraction, probabilistic forecasting, and risk-aware abstention.
2. A controlled evaluation on POPCAST across 5 systems, 10 tasks, and 130 total runs using Brier score, log loss, and task-success rate.
3. A detailed negative-result analysis showing that market-only forecasting outperforms full retrieval in aggregate (Brier 0.076 vs. 0.173), largely because retrieval amplifies rumor and leakage risk on hard celebrity and music tasks.
4. A set of concrete future directions: frozen evidence snapshots, finer-grained source credibility, abstention tuned by coverage-risk curves, and market-aware trading policies.

2 AGENT DESIGN

2.1 ARCHITECTURE OVERVIEW

KALIBRATE processes each market as an independent forecasting episode. The pipeline is shown in Figure 1.

KALIBRATE has five modules. The market-state constructor builds the normalized input seen by all systems: market title, domain, resolution criteria, decision timestamp, time to resolution, price history, volatility, liquidity, and context. Hidden outcome fields and post-decision notes are stripped. The tool router decides whether to search and can issue at most $B = 3$ Tavily queries. Retrieved documents are filtered against the decision timestamp before entering the model context. The evidence extractor converts remaining documents into typed signals such as relevance, source credibility, event type, sentiment, and recency. The forecaster outputs a schema-constrained probability in $[0.05, 0.95]$ plus confidence and a short reasoning summary. The risk controller can abstain by returning $p = 0.5$ when confidence is low, volatility is high, or evidence relevance is weak. Further implementation details and prompts appear in Appendix A.

3 EXPERIMENTAL SETUP

3.1 BENCHMARK

We evaluate on POPCAST, a 10-task historical benchmark of pop-culture prediction markets. Tasks span film, television, music, awards, celebrity activity, streaming, and viral media. Difficulty is distributed as 2 easy, 4 medium, and 4 hard tasks. Each task has a binary outcome $y \in \{0, 1\}$ and a decision timestamp that defines the information cutoff.

Table 1: Overall performance on POPCAST. Lower Brier and log loss are better; higher TSR is better. Mean $\pm$ standard deviation is across three trials for non-deterministic systems.

System	Brier $\downarrow$	Log Loss $\downarrow$	TSR $\uparrow$
Random prior	0.250	0.693	0.400
Single-prompt LLM	0.101 ± 0.002	0.344 ± 0.007	0.900
Tool-enabled frontier	0.112 ± 0.007	0.406 ± 0.041	0.900
Market-only ablation	0.076 ± 0.001	0.299 ± 0.003	1.000
KALIBRATE full	0.173 ± 0.006	0.536 ± 0.028	0.800

3.2 SYSTEMS COMPARED

We compare five systems:

- **Random prior:** always predicts $p = 0.5$.
- **Single-prompt LLM:** a frontier LLM sees the market state only, with no tools.
- **Tool-enabled frontier:** a frontier LLM can call the same Tavily-backed search tool up to three times, but does not use KALIBRATE’s structured extraction or risk controller.
- **Market-only ablation:** KALIBRATE with tool budget $B = 0$, forecasting only from market state.
- **KALIBRATE full:** the complete pipeline with routing, retrieval, extraction, forecasting, and risk control.

All non-deterministic systems are run for three trials per task. The random baseline is deterministic and run once. This yields 130 total result files.

3.3 METRICS

Our primary metric is Brier score:

$$\text{Brier}(p, y) = (p - y)^2.$$

Lower is better. We also report log loss:

$$- [y \log(p) + (1 - y) \log(1 - p)],$$

which punishes confident wrong forecasts more harshly, and task-success rate (TSR), which measures whether $p > 0.5$ matches the binary outcome. Brier and TSR can disagree: a forecast of 0.51 and 0.95 are both directionally correct for a YES outcome, but the latter is much better calibrated.

Because the benchmark has only 10 tasks, we avoid strong claims from asymptotic significance tests. The appropriate final-report comparison is a paired permutation test over per-task Brier differences for overall system comparisons. Per-domain and per-difficulty results have very small n , so we report them as exploratory effect-size evidence rather than definitive hypothesis tests.

4 RESULTS

4.1 OVERALL RESULTS

Table 1 is the core result. KALIBRATE beats the random prior, but loses to every non-random baseline. The market-only ablation is best overall, with Brier 0.076 compared to 0.173 for the full agent. This means the price history, market metadata, and context hint already contain strong signal in this task set. Retrieval is not automatically beneficial; in aggregate, it hurts.

4.2 DIFFICULTY AND DOMAIN PATTERNS

The aggregate result is driven by hard tasks. KALIBRATE is competitive on easy and medium tasks, but reaches hard-task Brier 0.377 versus 0.118 for market-only. Appendix E gives the full difficulty and per-task tables. Domain analysis also shows that market-only does not win uniformly:

the full agent beats market-only on film and ties on television, streaming, and viral. The overall gap is driven primarily by celebrity and music. On celebrity, market-only Brier is 0.072 while the full agent reaches 0.781 because retrieval pushes the model toward a confident YES on the Taylor Swift / Travis Kelce engagement task. On music, the full agent over-weights cultural momentum and streaming hype relative to chart-position constraints.

5 DISCUSSION AND FUTURE WORK

The market-only ablation won because it preserved the strong prior encoded in price history and context. Retrieval added marginal information, but also introduced noise. We identify three main failure modes: evidence over-weighting, where repeated documents moved the forecast too far from the prior; coarse source credibility, where entertainment outlets with very different reliability were treated similarly; and over-eager retrieval, where the router searched even when the market state was already strong. Qualitative examples appear in Appendix B.

The single-prompt and tool-enabled frontier baselines use a stronger frontier model than the default KALIBRATE components, so they are intentionally difficult comparisons. Historical pop-culture tasks also create possible training-data leakage: a frontier model may have internal knowledge of resolved events. This is why the matched comparison between full KALIBRATE and market-only KALIBRATE is especially important.

Future work should freeze retrieved evidence snapshots, improve source credibility modeling, tune abstention with coverage-risk curves, and move from one-shot forecasts toward market-aware position sizing. Additional limitations and details appear in Appendix C.

6 CONCLUSION

We presented KALIBRATE, a structured forecasting agent for pop-culture prediction markets. The system combines market-state construction, budgeted retrieval, evidence extraction, probability forecasting, and risk-aware abstention. On POPCAST, the full agent outperforms a random prior but underperforms stronger non-random baselines and its own market-only ablation.

The central lesson is that tools are not automatically helpful. In our benchmark, market priors were strong and retrieval was dangerous when evidence came from noisy entertainment domains. The path forward is not simply more search; it is better evidence auditing, better source credibility, calibrated abstention, and principled decisions about when the market prior should dominate.

A IMPLEMENTATION NOTES

KALIBRATE is implemented in Python. The default agent model is `gpt-4o-mini`; the single-prompt and tool-enabled frontier baselines use `gpt-4o`. Tavily is used for web search. Each full-agent episode can use up to three search calls, each returning up to five results before filtering and deduplication.

A.1 MODULE DETAILS

The market state constructor normalizes each task into a JSON object visible to all systems. It includes the market title, domain, resolution criteria, decision timestamp, resolution timestamp, time to resolution, recent price history, volatility proxy, liquidity proxy, and a context hint. In a live Kalshi copilot setting, this state would be built from the market page or API: current YES/NO price, recent price movement, liquidity, time until resolution, and visible market rules.

The tool router decides whether to invoke web search and, if so, which queries to issue. It returns an `invoke_tools` flag, a list of queries, and a rationale. Search results are filtered by publication timestamp before reaching downstream modules. Raw search results are then converted into structured evidence labels: sentiment, source credibility, event type, relevance score, and recency weight. The forecaster receives the market state and optional evidence summary, then outputs `p_hat`, confidence, signals used, and a short reasoning summary. The risk controller implements abstention as $p = 0.5$.

B QUALITATIVE EXAMPLES

Success case: The Weeknd Grammy nomination. On T6, the agent predicted that The Weeknd would not receive a Grammy nomination. This was correct. Retrieval helped because the relevant evidence was specific and tied to the resolution criterion: his public conflict with the Recording Academy and uncertainty around submission. The evidence extractor produced a negative directional score, and the final forecast remained below 0.5. This is the case KALIBRATE was designed for: dated, credible evidence that clarifies an ambiguous market.

Failure case: Taylor Swift / Travis Kelce engagement. T8 is the agent’s clearest failure. The outcome was NO, but KALIBRATE predicted around 0.90 YES. The retrieved evidence contained many documents about engagement rumors or later engagement-related information; the evidence extractor treated much of this as high relevance, and the forecaster interpreted repeated coverage as confirmation. The tool-enabled frontier baseline made a similar error. Market-only, by avoiding retrieval, stayed near 0.30 and scored well. This is both a model failure and a benchmark lesson: timestamp filtering is necessary but not sufficient if snippets, pages, or metadata can encode post-cutoff knowledge.

Mixed case: Charli XCX’s *Brat*. On T10, KALIBRATE’s raw forecast was directionally correct in one representative trial ($p_{\text{raw}} = 0.35$ for a NO outcome), but the risk controller abstained due to high volatility and returned 0.5. This produced Brier 0.25. Here the issue was not retrieval overconfidence but abstention calibration: the system correctly sensed uncertainty but erased a useful forecast. This motivates coverage-risk analysis, where we evaluate how Brier changes as the agent forecasts on fewer or more markets.

C LIMITATIONS AND FUTURE WORK DETAILS

The largest limitation is sample size: POPCAST has 10 tasks, so domain-level conclusions are exploratory. The second limitation is benchmark construction: fixed JSON tasks prevent outcome leakage through the task state, but live retrieval can still leak through snippets or metadata. The third limitation is abstention. A fixed volatility threshold caused KALIBRATE to abstain on T10 even when its raw probability was directionally useful. We plan to tune abstention with coverage-risk curves, where coverage is the fraction of markets forecasted and risk is mean Brier on that subset. Finally, a deployed market agent should move beyond one-shot forecasts toward position sizing and trading frequency based on confidence, market price, volatility, and new evidence.

D PROMPT SKETCH

D.1 ROUTER

Given a JSON market state, decide whether external retrieval is needed. If retrieval would help, output up to three concise search queries. Never request information after the decision timestamp.

D.2 FORECASTER

You are a calibrated probabilistic forecaster. Given a market state and optional evidence features, output only JSON with `p_hat`, `confidence`, `signals_used`, and `reasoning_summary`.

E ADDITIONAL RESULTS

Table 2: Brier score by difficulty. Lower is better.

System	Overall	Easy	Medium	Hard
Random prior	0.250	0.250	0.250	0.250
Single-prompt LLM	0.101	0.032	0.116	0.121
Tool-enabled frontier	0.112	0.004	0.006	0.273
Market-only ablation	0.076	0.049	0.046	0.118
KALIBRATE full	0.173	0.018	0.046	0.377

Table 3: Brier score by domain. Lower is better. Domain-level results are exploratory because several domains contain only one task.

System	Awards	Celebrity	Film	Music	Streaming	TV	Viral
Random prior	0.250	0.250	0.250	0.250	0.250	0.250	0.250
Single-prompt LLM	0.017	0.109	0.240	0.091	0.011	0.020	0.176
Tool-enabled frontier	0.007	0.872	0.004	0.046	0.007	0.005	0.124
Market-only ablation	0.023	0.072	0.100	0.071	0.023	0.023	0.250
KALIBRATE full	0.036	0.781	0.070	0.220	0.023	0.023	0.250

Table 4: Per-task Brier scores. Means are across trials where applicable.

Task	Domain	Diff.	Random	Single	Tools	Full
T1	Film	Easy	0.250	0.058	0.005	0.018
T2	Film	Medium	0.250	0.423	0.003	0.122
T3	TV	Medium	0.250	0.020	0.005	0.023
T4	Music	Hard	0.250	0.176	0.090	0.423
T5	Music	Easy	0.250	0.006	0.003	0.018
T6	Awards	Hard	0.250	0.022	0.005	0.055
T7	Awards	Medium	0.250	0.011	0.010	0.017
T8	Celebrity	Hard	0.250	0.109	0.872	0.781
T9	Streaming	Medium	0.250	0.011	0.007	0.023
T10	Viral	Hard	0.250	0.176	0.124	0.250